

Information-Theoretically Secure Aggregation for Lightweight Federated Learning: Resilient to Dropouts and Adversaries

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Abstract—On-device federated learning (FL) enables privacy-preserving and personalized model training on resource-constrained devices such as smartphones and IoT nodes. To reduce communication cost, sign-based methods (e.g., SIGNSGD) transmit one-bit gradients. However, exposing gradient signs makes them vulnerable to inference attacks, while existing secure aggregation schemes are often incompatible with such methods or incur significant computational and communication overhead. We propose a lightweight and information-theoretically secure aggregation framework tailored for sign-based FL. The framework securely computes the majority vote (MV) polynomial through single-round secure multiplication, ensuring end-to-end information-theoretic security under the honest-majority assumption while revealing only the final aggregated sign to the server. To enhance efficiency and scalability, we introduce two key techniques. First, inverse-form exponent reduction halves the effective MV polynomial degree, reducing both communication and computation costs. Second, we propose single-round secure multiplication, achieving linear offline complexity and storage with only a single online communication. Together, these techniques reduce online communication by up to 99.5% and latency by up to 85.7% compared to conventional approaches. Also, by leveraging inherent MDS-code-based decoding, the framework achieves robustness against both dropouts and adversarial behaviors, yielding accuracy gains of up to 20.65% and 10.74%, respectively. Overall, the proposed framework establishes a practical foundation for large-scale, low-latency, and information-theoretically secure aggregation in sign-based FL.

Index Terms—Dropouts and adversaries, information-theoretically secure aggregation, majority vote polynomial, sign-based federated learning, single-round multiplication.

I. INTRODUCTION

The proliferation of intelligent mobile devices and the Internet of Things (IoT) has accelerated the adoption of on-device federated learning (FL)—a decentralized machine learning paradigm that enables privacy-preserving model training directly on user devices [1]–[12]. By training locally without exposing raw data, FL addresses privacy concerns and is particularly attractive for resource-constrained environments such as smartphones, wearables, and edge networks.

Despite its advantages, communication efficiency remains a critical bottleneck in FL, as participating devices often operate under stringent bandwidth and computation constraints [13]. To alleviate this issue, gradient compression techniques such as SIGNSGD have been proposed, where model updates

are quantized into compact one-bit representations [14]–[19]. While this approach significantly reduces communication cost, it also exposes individual gradient signs, making sign-based methods inherently vulnerable to inference attacks and privacy leakage [20], [21]. Recent studies demonstrate that gradient signs can be exploited for gradient inversion and membership inference attacks [21], [22], highlighting the urgent need for end-to-end privacy protection even under extreme 1-bit quantization.

Existing secure aggregation schemes have attempted to mitigate these vulnerabilities, but significant limitations remain. Approaches based on homomorphic encryption introduce prohibitive computational and communication overheads, rendering them unsuitable for resource-constrained edge or IoT environments. In addition, protocols that rely on preprocessed correlated randomness—such as Beaver triples or other MPC-style offline preprocessing [23]—incur substantial precomputation, storage, and communication costs, which further increase protocol complexity and limit practicality on lightweight devices [24]. Furthermore, most conventional secure aggregation frameworks are inherently incompatible with sign-based methods due to their structural mismatch. In particular, prior schemes assume linear aggregation over multi-bit gradients, whereas sign-based methods require a secure nonlinear majority voting operation which cannot be efficiently supported by conventional frameworks.

A recent work, Hi-SAFE [25], presents the first secure aggregation framework *compatible with SIGNSGD* with majority vote (SIGNSGD-MV), providing only computational security under cryptographic hardness assumptions while maintaining simple server-side aggregation and low latency through hierarchical subgrouping. However, it still depends on preprocessed Beaver triples, incurring offline overhead, and its additive secret sharing makes it vulnerable to user dropouts and adversarial behaviors—an important limitation in practical FL settings.

To overcome these challenges, we focus on *information-theoretic security*, which guarantees privacy even against adversaries with unbounded computational power [26]. Unlike computational approaches, it is entirely independent of any computational hardness assumptions or heavy cryptographic mechanisms such as homomorphic encryption or correlated randomness. This unconditional protection relies solely on randomness and the algebraic properties of secret sharing, making it particularly suitable for lightweight and long-lived

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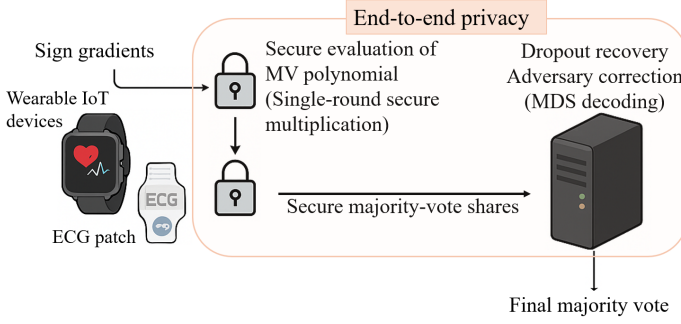


Fig. 1: Overview of the proposed FL framework. Sign gradients (1-bit each) are securely aggregated via single-round MV polynomial evaluation. Only the final majority vote is revealed to the server, ensuring end-to-end privacy with minimal communication overhead.

FL deployments where cryptographic assumptions or key management may not be practical.

Building upon this foundation, we present a *lightweight and information-theoretically secure aggregation framework* for sign-based FL. The core contribution lies in recognizing that the majority vote (MV) polynomial possesses a distinctive *single-base structure*—all monomials involve powers of a single aggregated value $\mathbf{x} = \sum_{i=1}^n \mathbf{x}_i$, where $\mathbf{x}_i$ is the one-bit gradient of user i , rather than products of distinct inputs. This structural insight enables a fundamentally more efficient secure multiplication approach.

We propose two complementary single-round secure multiplication methods. **Method A** (DN + Classical N -BTE) handles general N -input multiplication using an subset-mask table, suitable for small-scale settings. **Method B** (Single-Mask Power Sharing) exploits the single-base structure of MV polynomials: by employing a single random mask and the binomial expansion, offline computational complexity and storage are reduced from exponential to linear while requiring only a single online opening. Combined with the *inverse-form exponent reduction* that halves the polynomial degree, the proposed framework achieves up to **99.5% reduction in online communication** and **85.7% reduction in latency** compared to the conventional Beaver-triple-based approaches.

Furthermore, by leveraging inherent MDS-code-based decoding properties of the secure multiplication, the proposed framework attains strong robustness against both user dropouts and adversarial corruptions, **improving practical FL accuracy by up to 20.65% and 10.74% under dropout and adversarial conditions**, respectively. Fig. 1 illustrates the overall system, where one-bit local gradients are securely aggregated in a single round, revealing only the final MV result to the server and thus achieving full end-to-end privacy.

A. Contributions

The key contributions of this paper are summarized as follows:

- **Lightweight and information-theoretically secure aggregation.** We propose a lightweight secure aggregation

framework that ensures *information-theoretic privacy* under the honest-majority assumption, without relying on any computational hardness assumptions or heavy cryptographic mechanisms. It remains secure against computationally unbounded adversaries, making it suitable for long-lived, resource-constrained FL deployments.

- **End-to-end privacy via secure MV polynomial evaluation.** The framework securely evaluates MV polynomials for sign-based FL, revealing only the final aggregated sign to the server and preventing inference from intermediate gradient signs.
- **Inverse-form exponent reduction.** We introduce an algebraic transformation that rewrites high-degree terms as inverse powers, halving the maximum polynomial degree and further reducing both communication and computational costs.
- **Single-round secure multiplication.** We propose two single-round secure multiplication methods that enable secure evaluation of MV polynomials, thereby achieving end-to-end privacy:
 - (i) **Method A (DN + N -BTE):** A two-layer method where Layer 1 performs DN-style binary multiplications and Layer 2 aggregates results via the classical N -BTE. It achieves **single-round latency** with low online communication cost, but requires exponential offline computational complexity and storage.
 - (ii) **Method B (Single-Mask Power Sharing):** A structure-aware method that exploits the *single-base structure* of MV polynomials. Using a single mask and the binomial theorem, it achieves **single-round latency**, **linear** offline computational complexity and storage, with **minimal online communication**—only a single field element per user.
- **Robustness against both dropouts and adversarial behaviors.** By exploiting the inherent MDS-code properties of secure multiplication, the proposed framework achieves robustness against both user dropouts and malicious participants, achieving accuracy improvements of up to 20.65% and 10.74%, respectively.

B. Organization

The remainder of this paper is organized as follows. Section II reviews related work on secure aggregation and communication-efficient federated learning. Section III provides background on secure multiplication and recent advancements relevant to our work. Section IV presents the system model and problem formulation, and Section V details the proposed framework and its design components. Sections VI and VII provide theoretical analysis and experimental results, respectively, and Section VIII concludes the paper with discussions on practical implications and future research directions.

II. RELATED WORK

Secure aggregation has been widely investigated in FL. Masking-based protocols [23], [27] rely on pairwise secret sharing, but partial information may leak under semi-honest

TABLE I: Comparison of the proposed method with existing privacy-preserving aggregation approaches

Method	Security Model	Information Visible to Server	Accuracy Loss	Comm. Efficiency	Comp. Cost	Scalability	Dropout Resilience	Adversarial Robustness
Masking [23]	Computational (Double Masking)	Summation Values	No	Low	High	Limited	✓	✓
Local DP [30]	Information-Theoretic (ϵ -LDP)	Noisy Gradients	High	High	Low	High	✓	✗
HE [35]	Computational (RLWE-based HE)	No (Fully Encrypted)	No	Very Low	Very High	Very Limited	✗	✗
SIGNSGD-MV [14]	None	Raw Sign Gradients	No	Very High	Very Low	Very High	✗	✗
Hi-SAFE [25]	Computational (Beaver Triples)	Final Majority Vote Only	No	Medium	High	Medium	✗	✗
Proposed Method	Information-Theoretic (Single-Mask Power Sharing)	Final Majority Vote Only	No	High	Moderate	Very High	✓ (Recovery)	✓ (Correction)

settings. Differential privacy (DP) techniques [28]–[30] provide formal privacy guarantees, yet typically require strong noise that degrades accuracy. Homomorphic encryption (HE) approaches [31]–[37] achieve strong security, but they incur high computational cost and remain inefficient for nonlinear operations such as sign-based aggregation.

Communication-efficient schemes such as SIGNSGD and SIGNSGD-MV [14] reduce uplink traffic to one bit per parameter, but revealing raw gradient signs causes severe privacy leakage. Moreover, most existing secure aggregation frameworks are not directly compatible with such sign-based methods.

Hi-SAFE [25] is the first secure aggregation framework *compatible with* SIGNSGD-MV, providing computational security based on cryptographic hardness assumptions while maintaining simple server-side aggregation and low latency through hierarchical subgrouping. Nevertheless, it relies on preprocessed Beaver triples, which incur high offline overhead, and its additive secret sharing makes it vulnerable to user dropouts and adversarial behaviors—limitations that remain critical in practical FL environments.

In contrast, our work introduces a *lightweight and information-theoretically secure aggregation framework*, which removes heavy offline requirements while providing *information-theoretic security* under the honest-majority setting. The key insight is exploiting the *single-base structure* of MV polynomials, which enables linear offline computational complexity and storage—a dramatic improvement over exponential alternatives. Moreover, by leveraging the inherent *MDS-code* properties of the proposed secure multiplication, the framework attains robustness against both user dropouts and adversarial behaviors. Altogether, the proposed framework yields a more *scalable* and *resilient* foundation for secure aggregation in FL.

Comparison with Existing Secure Aggregation Methods

Table I provides a comparative summary of the proposed and existing secure aggregation methods in sign-based FL. The comparison considers multiple criteria including the type of security guarantee, exposure level to the server, accuracy preservation, and overall communication and computational efficiency.

As summarized, some existing methods such as masking [23] and local DP [30] offer partial protection and exhibit significant limitations when applied to sign-based protocols. In particular, masking-based approaches expose intermediate summation values during MV computation, and DP schemes suffer from accuracy degradation due to the addition of noise. While masking provides dropout resilience and adversarial robustness, it does not support full recovery of dropped shares or correction of corrupted values as our framework does—it only tolerates such failures without restoring the exact aggregation result. Similarly, DP offers dropout resilience but cannot correct adversarial corruptions. HE [35], while cryptographically strong and post-quantum secure, remains computationally intensive and is inefficient for nonlinear operations such as majority vote. Hi-SAFE [25] achieves computational security through Beaver-triple-based aggregation and reveals only the final majority vote to the server. However, it lacks dropout recovery and adversarial robustness.

In contrast, the proposed method achieves information-theoretic security under the honest-majority assumption and exploits the single-base structure of MV polynomials to achieve linear offline complexity and high communication efficiency. By leveraging the inherent MDS-code-based decoding, the framework enables *full recovery* of dropped user contributions and *complete correction* of adversarial corruptions within the decoding threshold, providing strong robustness without any accuracy degradation. Furthermore, all secure computations are performed in a single round, resulting in extremely low latency and high overall communication efficiency. Consequently, the proposed framework establishes a fully secure and resilient aggregation foundation for sign-based FL with information-theoretic privacy guarantees.

III. PRELIMINARIES

In this section, we briefly recall cryptographic building blocks relevant to our protocol design.

A. Beaver Triple Extension for N -Fan-in Multiplication

A Beaver triple [38] is a standard tool for secure MPC, enabling multiplication over secret-shared values without revealing private inputs. A Beaver triple $(\mathbf{a}, \mathbf{b}, \mathbf{c}) \in \mathbb{F}_p$ satisfies $\mathbf{c} = \mathbf{a} \cdot \mathbf{b} \pmod{p}$, where $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are secret-shared between two

Algorithm 1 N -Beaver Triple Extension (N -BTE) for N -Input Multiplication

- 1: **Input:** For each $\ell \in [N]$, parties P_0 and P_1 hold additive shares $\llbracket \mathbf{x}_\ell \rrbracket_0, \llbracket \mathbf{x}_\ell \rrbracket_1 \in \mathbb{F}_p$
 - 2: **Preprocessing:** Random masks $\{\mathbf{a}_\ell\}_{\ell=1}^N$; for all nonempty $I \subseteq [N]$, additive shares $\llbracket \mathbf{a}_I \rrbracket_0, \llbracket \mathbf{a}_I \rrbracket_1$ with $\mathbf{a}_I = \prod_{\ell \in I} \mathbf{a}_\ell$
 - 3: **Output:** Additive shares $\llbracket \mathbf{y} \rrbracket_0, \llbracket \mathbf{y} \rrbracket_1$ of $\mathbf{y} = \prod_{\ell=1}^N \mathbf{x}_\ell \bmod p$
 - 4: **Mask and Open:**
 - 5: **for** $k \in \{0, 1\}$ **do**
 - 6: **for** $\ell = 1$ **to** N **do**
 - 7: $P_k: \llbracket \hat{\mathbf{x}}_\ell \rrbracket_k \leftarrow \llbracket \mathbf{x}_\ell \rrbracket_k - \llbracket \mathbf{a}_\ell \rrbracket_k$
 - 8: Send $\llbracket \hat{\mathbf{x}}_\ell \rrbracket_k$ to the other party
 - 9: **end for**
 - 10: **end for**
 - 11: Reconstruct $\hat{\mathbf{x}}_\ell = \hat{\mathbf{x}}_\ell^{(0)} + \hat{\mathbf{x}}_\ell^{(1)} = \mathbf{x}_\ell - \mathbf{a}_\ell$ for all ℓ
 - 12: **Local Combination:**
 - 13: $P_0: \llbracket \mathbf{y} \rrbracket_0 \leftarrow \prod_{\ell=1}^N \hat{\mathbf{x}}_\ell + \sum_{\emptyset \neq I \subseteq [N]} \left(\prod_{\ell \notin I} \hat{\mathbf{x}}_\ell \right) \llbracket \mathbf{a}_I \rrbracket_0$
 - 14: $P_1: \llbracket \mathbf{y} \rrbracket_1 \leftarrow \sum_{\emptyset \neq I \subseteq [N]} \left(\prod_{\ell \notin I} \hat{\mathbf{x}}_\ell \right) \llbracket \mathbf{a}_I \rrbracket_1$
 - 15: **Correctness:** $\llbracket \mathbf{y} \rrbracket_0 + \llbracket \mathbf{y} \rrbracket_1 = \prod_{\ell=1}^N \mathbf{x}_\ell \bmod p$
 - 16: **Return:** $\llbracket \mathbf{y} \rrbracket_0, \llbracket \mathbf{y} \rrbracket_1$
-

parties P_0 and P_1 . Precomputed Beaver triples enable each *binary* secure multiplication to be performed in a single online round of interaction, after the correlated random Beaver triples $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ are sufficiently generated during the offline phase. For any value $\mathbf{x} \in \mathbb{F}_p$, we denote by $\llbracket \mathbf{x} \rrbracket = (\llbracket \mathbf{x} \rrbracket_0, \llbracket \mathbf{x} \rrbracket_1)$ its two-party additive sharing.

N -Beaver Triple Extension (N -BTE): To extend this idea to an N -input multiplication $\mathbf{z} = \prod_{\ell=1}^N \mathbf{x}_\ell$, the *N -Beaver Triple Extension (N -BTE)* [39] precomputes not only the N random masks $\{\mathbf{a}_\ell\}_{\ell=1}^N$ but also all subset multiplications

$$\mathbf{a}_I = \prod_{\ell \in I} \mathbf{a}_\ell, \quad I \subseteq [N], |I| \geq 2,$$

each of which is secret-shared between P_0 and P_1 and $[N] \triangleq \{1, 2, \dots, N\}$. During the online phase, both parties jointly mask each shared input $\mathbf{x}_\ell$ and open the masked difference $\hat{\mathbf{x}}_\ell = \mathbf{x}_\ell - \mathbf{a}_\ell$, so that the values $\{\hat{\mathbf{x}}_\ell\}$ become public. Then, using the precomputed subset shares $\{\mathbf{a}_I\}$, both parties locally reconstruct additive shares of the multiplication. Correctness follows from the algebraic identity

$$\prod_{\ell=1}^N (\hat{\mathbf{x}}_\ell + \mathbf{a}_\ell) = \sum_{I \subseteq [N]} \left(\prod_{\ell \notin I} \hat{\mathbf{x}}_\ell \right) \mathbf{a}_I = \prod_{\ell=1}^N \mathbf{x}_\ell \pmod{p},$$

which ensures that the final reconstructed value equals the true N -input multiplication [39].

Remark: N -BTE achieves *constant-round communication independent of N* , greatly reducing latency compared with sequential binary multiplications. However, preprocessing and storage costs scale exponentially as 2^N , since all subset multiplications $\{\mathbf{a}_I\}$ must be prepared offline. In practice, N is therefore limited (typically $N \leq 8-9$) to balance offline cost against one-round online efficiency.

B. ATLAS Multiplication

ATLAS [40] is an information-theoretically secure MPC protocol that improves both communication and round complexities under the honest-majority assumption ($t < n/2$), where t is the maximum number of colluding or corrupted parties. While DN method [26] is simple and robust, its round and communication costs limit scalability for deep multiplication chains.

ATLAS introduces two orthogonal optimizations:

- *Communication reduction.* By using t -wise independent randomness, only t random double sharings need to be uniformly random, while the rest are derived via a hyper-invertible matrix. This amortizes the preprocessing cost and reduces total communication cost.
- *Round reduction.* By forcing an output of DN multiplication into a Beaver-triple-friendly form, ATLAS allows Beaver triples for the next layer to be prepared in parallel. As a result, two consecutive layers of parallel multiplications can be evaluated in one reconstruction round, halving the number of rounds.

Two-Layer Evaluation: ATLAS efficiently evaluates two layers of multiplications by combining DN-style masking in the first layer with binary Beaver triples in the second layer as in Algorithm 2. Here, $[\mathbf{x}]_t$ denotes t -sharing of $\mathbf{x} \in \mathbb{F}_p$, and $[\mathbf{x}]_{2t}$ denotes $2t$ -sharing of the same value $\mathbf{x}$.

Algorithm 2 ATLAS Two-Layer Multiplication

- 1: **Input:** Degree- t sharings $[\mathbf{x}]_t, [\mathbf{y}]_t$
 - 2: **Output:** Degree- t sharings of output $[\mathbf{z}]_t$ for $\mathbf{z} = \mathbf{x} \cdot \mathbf{y}$
 - 3: **Preprocessing:**
 - 4: Generate double sharings $([\mathbf{r}_1]_t, [\mathbf{r}_1]_{2t}), ([\mathbf{r}_2]_t, [\mathbf{r}_2]_{2t})$
 - 5: Set $[\mathbf{a}]_t \leftarrow [\mathbf{r}_1]_t, [\mathbf{b}]_t \leftarrow [\mathbf{r}_2]_t$
 - 6: Compute $[\mathbf{c}]_t \leftarrow [\mathbf{a} \cdot \mathbf{b}]_t$ via secure multiplication
 - 7: **First Layer (DN):**
 - 8: $[\mathbf{e}_1]_{2t} \leftarrow [\mathbf{x}]_t \cdot [\mathbf{y}]_t + [\mathbf{r}_1]_{2t}$; Open to get $\mathbf{e}_1$
 - 9: $[\mathbf{e}_2]_{2t} \leftarrow [\mathbf{x}]_t \cdot [\mathbf{y}]_t + [\mathbf{r}_2]_{2t}$; Open to get $\mathbf{e}_2$
 - 10: **Second Layer (Beaver):**
 - 11: Compute $[\mathbf{z}]_t \leftarrow \mathbf{e}_1 \mathbf{e}_2 - \mathbf{e}_1 [\mathbf{b}]_t - \mathbf{e}_2 [\mathbf{a}]_t + [\mathbf{c}]_t$
 - 12: **Return:** $[\mathbf{z}]_t$
-

Remark: ATLAS reduces online communication cost and latency by merging two consecutive multiplication layers into a single reconstruction round. This optimization is effective for latency-sensitive workloads but, in its canonical form, ATLAS supports only binary multiplications. Therefore, it does not achieve one-round aggregation for general N -input multiplications. This limitation motivates our Method A (Section V-B1), which extends the ATLAS two-layer structure to support N -input multiplication by integrating the classical N -BTE in the second layer.

C. Construction of MV Polynomials via Fermat's Little Theorem

Fermat's Little Theorem is a cornerstone of number theory with significant implications in cryptography, modular arithmetic, and secure computation [41]. It provides an efficient

means to simplify large exponentiations modulo a prime, which is particularly advantageous for constructing indicator functions over finite fields.

Indicator Function via Fermat's Little Theorem: Formally, if p is a prime and a is an integer such that $a \not\equiv 0 \pmod{p}$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

From this, an indicator function over $\mathbb{F}_p$ can be derived: for $x, a \in \mathbb{F}_p$,

$$\mathbb{1}_{\{x=a\}} = 1 - (x - a)^{p-1} \pmod{p} = \begin{cases} 1, & \text{if } x = a, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

This compact formulation is particularly useful for mapping discrete decision rules into polynomial form, enabling efficient precomputation and evaluation in secure majority-vote aggregation [25].

Formulation of MV Polynomials: Using Eq. (1), we can construct an MV polynomial for the aggregated vector $\mathbf{x} = \sum_{i=1}^n \mathbf{g}_i \in \mathbb{F}_p^d$, where $\mathbf{g}_i \in \{-1, +1\}^d$ is the locally updated gradient sign of user i and d is the model dimension. Let p be the smallest prime greater than n , where n is the number of active users, ensuring correctness of the finite-field indicator mapping. Since the majority vote is applied independently to each coordinate j , the MV polynomial is defined component-wise as:

$$F(\mathbf{x}) = (F(x^{(1)}), F(x^{(2)}), \dots, F(x^{(d)})),$$

where $x^{(j)} = \sum_{i=1}^n g_i^{(j)}$.

Tie-breaking Policy: When n is even, a tie ($\mathbf{x} = 0$) may occur, and the tie-breaking policy must be defined:

- $\text{sign}(0) \in \{-1, +1\}$: tie resolved to a binary decision (1-bit output of $F(\mathbf{x})$),
- $\text{sign}(0) = 0$: tie represented as a distinct third state (2-bit output of $F(\mathbf{x})$).

This policy impacts both the structure of $F(\mathbf{x})$ and the communication bandwidth requirements.

Final MV Polynomials: For a scalar input $x = \sum_{i=1}^n x_i$ with $x_i \in \{-1, +1\}$, the MV polynomial for computing the MV of $x_1, x_2, \dots, x_n$ is given by [25]

$$F(x) = \sum_{m \in \{-n, -n+2, \dots, n\}} \text{sign}(m) [1 - (x - m)^{p-1}] \pmod{p}, \quad (2)$$

where p is the smallest prime greater than n , $m = \sum_{i=1}^n m_i$ with $m_i \in \{-1, +1\}$, and $\text{sign}(0)$ is determined by the tie-breaking rule.

Remark: Once n and the tie-breaking rule are fixed, $F(\mathbf{x})$ can be systematically constructed and precomputed according to Eq. (2), allowing efficient MV evaluation in secure aggregation protocols. Table II lists representative examples of precomputed MV polynomials according to tie-breaking policies.

IV. SYSTEM MODEL AND PROBLEM FORMULATION

This section provides an overview of the FL framework and formulates the system model used throughout this work, with a focus on the SIGNSGD-MV update mechanism.

TABLE II: Precomputed MV polynomials $F(\mathbf{x})$

#Users	$\text{sign}(0) \in \{-1, +1\}$	$\text{sign}(0) = 0$
$n = 3$	$2\mathbf{x}^3 + 4\mathbf{x} \pmod{5}$	$2\mathbf{x}^3 + 4\mathbf{x} \pmod{5}$
$n = 4$	$\mathbf{x}^4 + 3\mathbf{x}^3 + \mathbf{x} + 4 \pmod{5}$	$3\mathbf{x}^3 + \mathbf{x} \pmod{5}$
$n = 5$	$3\mathbf{x}^5 + 2\mathbf{x}^3 + 3\mathbf{x} \pmod{7}$	$3\mathbf{x}^5 + 2\mathbf{x}^3 + 3\mathbf{x} \pmod{7}$
$n = 6$	$\mathbf{x}^6 + 4\mathbf{x}^5 + 5\mathbf{x}^3 + 4\mathbf{x} + 6 \pmod{7}$	$4\mathbf{x}^5 + 5\mathbf{x}^3 + 4\mathbf{x} \pmod{7}$

A. FL Setup and Review

The goal of FL is to collaboratively optimize a shared model parameter $\theta \in \mathbb{R}^d$ across a set of distributed users without revealing their raw data. The optimization problem is formulated as:

$$\theta^* = \arg \min_{\theta \in \mathbb{R}^d} f(\theta) = \arg \min_{\theta \in \mathbb{R}^d} \sum_{i=1}^{N_t} \frac{m_i}{m} f_i(\theta), \quad (3)$$

where N_t denotes the total number of users, m_i is the number of data samples held by user i , and $m = \sum_{i=1}^{N_t} m_i$ is the total number of samples. Letting $D_{i,j}$ be the j -th data sample of user i , the local loss function of user i , denoted by $f_i(\theta)$, is defined as

$$f_i(\theta) = \frac{1}{m_i} \sum_{j=1}^{m_i} L(\theta; D_{i,j}), \quad (4)$$

where $L(\theta; D)$ is the loss of a model θ on an input-output pair D (e.g., least-square loss, cross-entropy loss, etc.).

1) *Standard FL Framework:* Typical FL algorithms proceed in synchronized communication rounds, consisting of local updates followed by global aggregation. In each global round τ , the server randomly selects a subset $\mathcal{S}_\tau \subseteq [N_t]$ of n active users [42]. Let $C = \frac{n}{N_t}$ be the ratio of active users who participate in each global round. Each active user performs local updates with its own data and sends the updated model back to the server. The de-facto algorithm of FL is FedAvg [42], which operates as follows:

- (Downlink transmission) The server samples n active users from all N_t users (i.e., $\mathcal{S}_\tau \subseteq [N_t]$ with $|\mathcal{S}_\tau| = n$) and broadcasts the current model $\theta(\tau)$ to them.
- (Local update) Each user $i \in \mathcal{S}_\tau$ sets $\theta_i(\tau) = \theta(\tau)$ and updates its local model via SGD: $\theta_i(\tau+1) = \theta(\tau) - \eta \mathbf{g}_i(\tau)$, where $\mathbf{g}_i(\tau) = \nabla f_i(\theta(\tau))$ and $\eta > 0$ is the learning rate.
- (Uplink transmission) Each user $i \in \mathcal{S}_\tau$ sends the updated local information $\theta_i(\tau+1)$ to the server.
- (Global update) From the aggregated information $\{\theta_i(\tau+1) : i \in \mathcal{S}_\tau\}$, the server updates the global model via the weighted average: $\theta(\tau+1) = \frac{\sum_{i \in \mathcal{S}_\tau} m_i \theta_i(\tau+1)}{\sum_{i \in \mathcal{S}_\tau} m_i}$.

B. Communication-Efficient FL via SIGNSGD-MV

As FL models grow in size, communication becomes a critical bottleneck. To mitigate this, the SIGNSGD-MV approach [14], [16] reduces uplink communication by applying 1-bit quantization (i.e., sending $\text{sign}(\mathbf{g}_i(\tau)) \in \{-1, +1\}^d$) and reduces downlink communication by sending back a majority vote of these signs. This compresses communication in both directions. The overall process is summarized in Algorithm 3. In this work, we build upon this protocol to design a secure and communication-efficient aggregation framework suited for resource-constrained environments such as IoT.

Algorithm 3 SIGNSGD-MV Protocol [14]

- 1: **Input:** Learning rate η , current model $\theta(\tau)$, #active users n , and local gradients $\mathbf{g}_i(\tau)$
 - 2: **On Server:**
 - 3: Pull $\text{sign}(\mathbf{g}_i(\tau))$ from each user
 - 4: Push $\text{sign}[\sum_{i=1}^n \text{sign}(\mathbf{g}_i(\tau))]$ to each user
 - 5: **On User:**
 - 6: $\theta(\tau + 1) \leftarrow \theta(\tau) - \eta \cdot \text{sign}[\sum_{i=1}^n \text{sign}(\mathbf{g}_i(\tau))]$
-

C. System Assumptions and Problem Scope

We consider an FL environment under the following assumptions:

- **Semi-Honest Model:** All users follow the protocol correctly, but up to t of them may collude to infer private information from the exchanged messages [23], [43]–[48].
- **SIGNSGD-MV Update Rule:** Each of n users transmits a 1-bit sign of its local gradient, and the server computes the majority vote over the received signs to determine the update direction [14]–[18].
- **Honest-Majority Setting:** Information-theoretic security requires an honest majority, i.e., $t < n/2$. This assumption naturally aligns with SIGNSGD-MV, which relies on the majority of users providing correct local signs; if $t \geq n/2$, the global majority vote itself becomes incorrect. Thus, the honest-majority setting is both a natural and essential condition for ensuring correctness and privacy.

These assumptions provide the foundation for our proposed secure aggregation framework. In the following section, we present two complementary methods that achieve information-theoretic security under these assumptions while minimizing online communication costs and latency.

V. PROPOSED SECURE AGGREGATION METHODS

This section presents two complementary components that together enable efficient and information-theoretically secure evaluation of MV polynomials in FL, ensuring end-to-end privacy by revealing only the final aggregated sign to the server. Unlike prior approaches that rely solely on binary Beaver triples or DN-style multiplication [26], [38], [40], the proposed framework introduces new primitives that jointly achieve substantial reduction in *communication cost* and *latency*, while keeping preprocessing and storage manageable.

- (i) **Inverse-Form Exponent Reduction.** We introduce an inverse-form polynomial construction that *halves the maximum exponentiation depth*. This transformation reduces the number and depth of secure multiplications required to evaluate the MV polynomial, leading to lower communication and computational costs without affecting accuracy or security. The reduced degree also enables scalable, single-round secure polynomial evaluation.
- (ii) **Single-Round Secure Multiplication.** To realize one-round secure multiplications, we propose two realizations, Method A and Method B, under a unified DN-based architecture, each targeting different multiplication structures:

- **Method A (DN + N-BTE).** This method adopts a two-layer structure: Layer 1 performs DN-style masked openings for binary multiplications, and Layer 2 aggregates the results using the classical N-BTE, enabling secure computation of the product of all pairwise multiplications within one round under an honest-majority setting. However, it requires an **exponential subset-mask table** of size $2^N - N - 1$, resulting in $\Theta(N \cdot 2^N)$ offline DN multiplications and $O(2^N)$ storage. While Method A minimizes online latency, its scalability is limited by the exponential offline cost.
- **Method B (Single-Mask Power Sharing).** For MV polynomial evaluation, Method B exploits the special structure where all multiplications involve powers of a **single aggregated value** $\mathbf{x}^k$ rather than products of distinct inputs. Using a *single mask* $\mathbf{a}$ and the binomial theorem, Method B achieves **linear** $O(N')$ offline computation and storage, where $N' = (p-1)/2$ is the maximum polynomial degree after inverse-form exponent reduction. This approach preserves the single-round online phase and honest-majority security while dramatically reducing overhead compared to Method A.

Next, a secure aggregation framework is proposed, which combines (i) inverse-form exponent reduction for lower multiplicative depth and (ii) single-round secure multiplication tailored to MV polynomial evaluation. Together, they achieve efficient, one-round, and information-theoretically secure aggregation suitable for large-scale federated learning.

A. Inverse-Form Exponent Reduction of MV Polynomials

MV polynomials recently arise in secure aggregation protocols for FL [25]. A representative construction, as in Eq. (2), involves high-degree monomials over a prime field $\mathbb{F}_p$, resulting in significant overhead for secure exponentiation.

Although such polynomials often contain monomials with degree $p-1$, Fermat's Little Theorem enables a simplification:

$$\mathbf{x}^{p-1} \equiv 1 \pmod{p} \quad \text{for all } \mathbf{x} \neq 0.$$

Then, the MV polynomial can be restructured into a mixed-degree polynomial:

$$F(\mathbf{x}) = \sum_{j=0}^{p-2} c_j \mathbf{x}^j \pmod{p}.$$

Also, high-degree terms $\mathbf{x}^j$ with $j > (p-1)/2$ can be algebraically reduced via the inverse-form transformation:

$$\mathbf{x}^j \equiv \mathbf{x}^{-(p-1-j)} \pmod{p}. \quad (5)$$

This transformation preserves correctness while effectively reducing the exponentiation depth.

Theorem 1 (Inverse-Form Polynomial Optimization). *Let $F(\mathbf{x}) = \sum_{j=0}^{p-2} c_j \mathbf{x}^j \in \mathbb{F}_p[\mathbf{x}]$ be a polynomial of degree at most $p-2$. Then, for all $j > (p-1)/2$, the following identity holds:*

$$c_j \mathbf{x}^j \equiv c_j \mathbf{x}^{-(p-1-j)} \pmod{p}.$$

Thus, $F(\mathbf{x})$ can be equivalently rewritten using terms of degrees $\leq (p-1)/2$ and their inverses.

Proof. It is trivial for $\mathbf{x} = 0$. Since Fermat's Little Theorem guarantees that $\mathbf{x}^{p-1} \equiv 1 \pmod{p}$ for all $\mathbf{x} \neq 0$, multiplying both sides by $\mathbf{x}^{j-(p-1)}$ yields $\mathbf{x}^j \equiv \mathbf{x}^{-(p-1-j)} \pmod{p}$. $\square$

Example (with $n = 5, p = 7$): Given $F(\mathbf{x}) = 3\mathbf{x}^5 + 2\mathbf{x}^3 + 3\mathbf{x}$, the inverse-form reduction yields:

$$F(\mathbf{x}) \equiv 3\mathbf{x}^{-1} + 2\mathbf{x}^3 + 3\mathbf{x} \pmod{7}.$$

Algorithm 4 shows the inverse-form reduction by rewriting all exponents $j > (p-1)/2$ in their negative inverse form.

Algorithm 4 Inverse-Form Exponent Reduction

- 1: **Input:** Coefficients $\{c_\ell\}_{\ell=1}^k$ with the corresponding exponents $\{j_\ell\}_{\ell=1}^k$ (for $F(\mathbf{x}) = \sum_{\ell=1}^k c_\ell \mathbf{x}^{j_\ell}$), modulus p
 - 2: **for** $\ell = 1$ **to** k **do**
 - 3: **if** $j_\ell > (p-1)/2$ **then**
 - 4: $j_\ell \leftarrow -(p-1-j_\ell)$
 - 5: **end if**
 - 6: **end for**
 - 7: **Output:** Transformed exponent-coefficient pairs $\{(c_\ell, j_\ell)\}_{\ell=1}^k$ (for $F(\mathbf{x}) = \sum_{\ell=1}^k c_\ell \mathbf{x}^{j_\ell}$)
-

In our setting, the modulus p is a small prime. Hence, modular inversion via Fermat's identity $\mathbf{x}^{-1} \equiv \mathbf{x}^{p-2} \pmod{p}$ is computationally efficient. However, for a large modulus or polynomial inversion over modular ring, Newton's method [49], [50] provides faster convergence and compatibility with FFT-based techniques. Thus, while the Fermat approach suffices in this work, Newton-based inversion may be adopted in future large-scale implementations. Hereafter, we always apply the inverse-form reduction to MV polynomials, so that the effective maximum degree is $N' = (p-1)/2$.

B. Single-Round Secure Multiplication

To enable efficient secure polynomial evaluation, we propose two one-round realizations—Method A and Method B—that achieve end-to-end information-theoretic security while targeting different multiplication structures.

We first present Method A as a general-purpose approach that handles arbitrary N -input multiplication, then introduce Method B as a specialized optimization for the MV polynomial evaluation. This progression clarifies why exploiting the single-base structure of MV polynomials yields dramatic efficiency gains: while Method A requires exponential offline complexity, Method B achieves linear complexity by recognizing that all monomials share an identical base value.

1) *Method A (DN + N-BTE):* Method A adopts a two-layer structure inspired by ATLAS: layer 1 performs DN-style masked openings for binary multiplications (feasible under $t < n/2$), producing outputs in Beaver-triple friendly form; layer 2 aggregates N outputs from layer 1 into one t -sharing *without additional openings* using precomputed subset products of the masks. This method computes $\prod_{\ell=1}^N (\mathbf{x}_\ell \cdot \mathbf{y}_\ell)$, the product of N binary multiplication results from layer 1 as follows:

Layer 1 (DN; one opening per binary): For each binary multiplication $[\mathbf{x}]_t \cdot [\mathbf{y}]_t$ with preprocessed pair of random double sharings $([\mathbf{a}]_t, [\mathbf{a}]_{2t})$ of the same secret $\mathbf{a}$:

- 1) Compute $[\mathbf{e}]_{2t} = [\mathbf{x}]_t \cdot [\mathbf{y}]_t + [\mathbf{a}]_{2t}$
- 2) Open $[\mathbf{e}]_{2t}$ to obtain public value $\mathbf{u} = \mathbf{x} \cdot \mathbf{y} + \mathbf{a}$
- 3) Set output sharing $[\mathbf{w}]_t = \mathbf{u} - [\mathbf{a}]_t$

The output is in *Beaver-triple friendly form*: $[\mathbf{w}]_t = \mathbf{u} - [\mathbf{a}]_t$ with public $\mathbf{u}$ and preprocessed $[\mathbf{a}]_t$.

Layer 2 (N-BTE; no extra opening): *Offline* (no openings; $t < n/2$): Given N base masks $\{[\mathbf{a}_\ell]_t\}_{\ell=1}^N$ from double sharings, generate all subset products $[\mathbf{a}_I]_t = \prod_{\ell \in I} [\mathbf{a}_\ell]_t$ for $I \subseteq [N], |I| \geq 2$, via DN with degree reduction, yielding $2^N - N - 1$ sharings.

Online: Given Beaver-triple friendly sharings $\{[\mathbf{w}_\ell]_t = \mathbf{u}_\ell - [\mathbf{a}_\ell]_t\}_{\ell=1}^N$ with public $\mathbf{u}_\ell$, compute t -shares of $\mathbf{z} = \prod_{\ell=1}^N (\mathbf{x}_\ell \cdot \mathbf{y}_\ell)$:

$$[\mathbf{z}]_t = \prod_{\ell=1}^N \mathbf{u}_\ell + \sum_{\emptyset \neq I \subseteq [N]} \alpha_I [\mathbf{a}_I]_t, \quad \alpha_I = (-1)^{|I|} \prod_{\ell \in [N] \setminus I} \mathbf{u}_\ell. \quad (6)$$

The overall protocol is summarized in Algorithm 5.

Complexity: $\Theta(N \cdot 2^N)$ DN multiplications and storage $2^N - N - 1$ for offline; N openings in one round for online.

Algorithm 5 DN + N-BTE (Method A)

- 1: **Input:** Degree- t sharings $\{([\mathbf{x}_\ell]_t, [\mathbf{y}_\ell]_t)\}_{\ell=1}^N$ for N binary multiplications
 - 2: **Output:** Degree- t sharing $[\mathbf{z}]_t$ of $\mathbf{z} = \prod_{\ell=1}^N (\mathbf{x}_\ell \cdot \mathbf{y}_\ell)$
 - 3: **Offline (input-independent; requires $t < n/2$):**
 - 4: Prepare random double sharings $\{([\mathbf{a}_\ell]_t, [\mathbf{a}_\ell]_{2t})\}_{\ell=1}^N$
 - 5: Generate $[\mathbf{a}_I]_t \leftarrow \prod_{\ell \in I} [\mathbf{a}_\ell]_t$ for all $I \subseteq [N]$ with $|I| \geq 2$ via DN multiplication { $2^N - N - 1$ items}
 - 6: **Online (Two-Layer Multiplication):**
 - 7: **Layer 1 (DN; one opening per binary):**
 - 8: **for** $\ell = 1$ **to** N **do**
 - 9: $[\mathbf{e}_\ell]_{2t} \leftarrow [\mathbf{x}_\ell]_t \cdot [\mathbf{y}_\ell]_t + [\mathbf{a}_\ell]_{2t}$
 - 10: Open $[\mathbf{e}_\ell]_{2t}$ to obtain public $\mathbf{u}_\ell = \mathbf{x}_\ell \cdot \mathbf{y}_\ell + \mathbf{a}_\ell$
 - 11: Set $[\mathbf{w}_\ell]_t \leftarrow \mathbf{u}_\ell - [\mathbf{a}_\ell]_t$ {Beaver-triple friendly form}
 - 12: **end for**
 - 13: **Layer 2 (N-BTE; no extra openings):**
 - 14: $[\mathbf{z}]_t \leftarrow \prod_{\ell=1}^N \mathbf{u}_\ell$ {public value}
 - 15: **for** each nonempty $I \subseteq [N]$ **do**
 - 16: $\alpha_I \leftarrow (-1)^{|I|} \cdot \prod_{\ell \in [N] \setminus I} \mathbf{u}_\ell$ {public coefficient}
 - 17: $[\mathbf{z}]_t \leftarrow [\mathbf{z}]_t + \alpha_I \cdot [\mathbf{a}_I]_t$
 - 18: **end for**
 - 19: **Return:** $[\mathbf{z}]_t$ { $\mathbf{z} = \prod_{\ell=1}^N (\mathbf{x}_\ell \cdot \mathbf{y}_\ell)$ }
-

Correctness: Layer 1: By the DN step with double-sharing $([\mathbf{a}_\ell]_t, [\mathbf{a}_\ell]_{2t})$ of the same secret $\mathbf{a}_\ell$, we have

$$\mathbf{w}_\ell = \mathbf{u}_\ell - \mathbf{a}_\ell = (\mathbf{x}_\ell \cdot \mathbf{y}_\ell + \mathbf{a}_\ell) - \mathbf{a}_\ell = \mathbf{x}_\ell \cdot \mathbf{y}_\ell.$$

Thus, $[\mathbf{w}_\ell]_t = \mathbf{u}_\ell - [\mathbf{a}_\ell]_t$ is a valid degree- t sharing of $\mathbf{x}_\ell \cdot \mathbf{y}_\ell$.

Layer 2: The N -input product is computed via the binomial-like expansion:

$$\prod_{\ell=1}^N \mathbf{w}_\ell = \prod_{\ell=1}^N (\mathbf{u}_\ell - \mathbf{a}_\ell)$$

$$\begin{aligned}
&= \sum_{I \subseteq [N]} (-1)^{|I|} \prod_{\ell \in [N] \setminus I} \mathbf{u}_\ell \cdot \prod_{\ell \in I} \mathbf{a}_\ell \\
&= \prod_{\ell=1}^N \mathbf{u}_\ell + \sum_{\emptyset \neq I \subseteq [N]} \alpha_I \cdot \mathbf{a}_I,
\end{aligned}$$

where $\alpha_I = (-1)^{|I|} \prod_{\ell \in [N] \setminus I} \mathbf{u}_\ell$ and $\mathbf{a}_I = \prod_{\ell \in I} \mathbf{a}_\ell$. Therefore, the output $[\mathbf{z}]_t$ is a correct degree- t sharing of $\prod_{\ell=1}^N (\mathbf{x}_\ell \cdot \mathbf{y}_\ell)$. Note that when $|I| = 1$, we have $\mathbf{a}_I = \mathbf{a}_\ell$, which is the base mask already available from the double sharings. Thus, only the subset products with $|I| \geq 2$ need to be precomputed offline.

Application to MV Polynomial Evaluation: When applied to MV polynomial evaluation with inverse-form exponent reduction, Method A computes each monomial $\mathbf{x}^k$ as a product of k formally distinct copies: $\mathbf{x}^k = \prod_{\ell=1}^k \mathbf{x}_\ell$ where $\mathbf{x}_\ell = \mathbf{x}$ for all $\ell \in [k]$. In this case, Layer 1 applies with $\mathbf{y}_\ell = 1$: $[\mathbf{e}_\ell]_{2t} = [\mathbf{x}]_t \cdot [1]_t + [\mathbf{a}_\ell]_{2t}$, $\mathbf{u}_\ell = \mathbf{x} + \mathbf{a}_\ell$, $[\mathbf{w}_\ell]_t = \mathbf{u}_\ell - [\mathbf{a}_\ell]_t$. Here $[1]_t$ denotes a degree- t sharing of the constant 1, and $[\mathbf{x}]_t \cdot [1]_t$ is computed via local component-wise multiplication, yielding a degree- $2t$ sharing of $\mathbf{x}$. Layer 2 then applies k -BTE with precomputed subset products $\{[\mathbf{a}_I]_t\}_{|I| \geq 2}$ to obtain $[\mathbf{x}^k]_t$. This requires $2^k - k - 1$ subset-mask sharings per monomial degree k , resulting in total offline complexity of $O(2^{N'})$, where $N' = (p-1)/2$ after inverse-form exponent reduction. Once all monomial sharings are obtained, the final output is:

$$[F(\mathbf{x})]_t = \sum_{j=0}^{N'} c_j \cdot [\mathbf{x}^j]_t. \quad (7)$$

While the online phase requires $O(N')$ openings, the exponential offline computational complexity and storage renders Method A impractical for large polynomial degrees. This limitation motivates Method B, which exploits the single-base structure to achieve linear computational complexity and storage.

2) Method B (Single-Mask Power Sharing for MV Polynomial): While Method A provides a general N -input multiplication using the classical subset-table approach, Method B offers a significantly more efficient alternative specifically tailored for MV polynomial evaluation. In the MV polynomial $F(\mathbf{x}) = \sum_{j=0}^{N'} c_j \mathbf{x}^j$ (where $N' = (p-1)/2$ after inverse-form reduction), each monomial $\mathbf{x}^k$ involves powers of a *single* aggregated value $\mathbf{x} = \sum_{i=1}^n \mathbf{g}_i$, rather than products of distinct inputs. This special structure enables a fundamentally simpler approach using a *single mask* $\mathbf{a}$ and the binomial theorem, reducing storage from exponential to linear while preserving one-round latency.

Key observation (Single-base exponentiation): For MV polynomial evaluation, we compute

$$\mathbf{x}^k = \underbrace{\mathbf{x} \cdot \mathbf{x} \cdot \mathbf{x} \cdots \mathbf{x}}_{k \text{ times}},$$

where all k factors are the *identical* base value $\mathbf{x}$. This admits the standard binomial expansion with a single mask $\mathbf{a}$:

$$\mathbf{x}^k = (\hat{\mathbf{x}} + \mathbf{a})^k = \sum_{r=0}^k \binom{k}{r} \hat{\mathbf{x}}^r \cdot \mathbf{a}^{k-r}, \quad (8)$$

where $\hat{\mathbf{x}} = \mathbf{x} - \mathbf{a}$ is the single masked value opened during the online phase.

Offline preparation (input-independent; $t < n/2$ required for degree reduction): For each aggregation round, sample a random mask $[\mathbf{a}]_t$ and compute all power sharings via sequential DN multiplication:

$$\begin{aligned}
&[\mathbf{a}^0]_t \leftarrow [1]_t, [\mathbf{a}^1]_t \leftarrow [\mathbf{a}]_t, \\
&[\mathbf{a}^j]_{2t} \leftarrow [\mathbf{a}^{j-1}]_t \cdot [\mathbf{a}]_t, [\mathbf{a}^j]_t \leftarrow \text{DegRed}([\mathbf{a}^j]_{2t}), j=2, \dots, N',
\end{aligned}$$

where **DegRed** denotes the degree reduction. This preparation requires $N' - 1$ DN multiplications with degree reduction. Since these operations involve only random masks (not actual inputs), they can be performed entirely offline without any openings, and $N' + 1$ degree- t sharings $\{[\mathbf{a}^j]_t\}_{j=0}^{N'}$ are stored.

Additionally, precompute the coefficient-weighted power sharings for efficient online evaluation:

$$[P_r]_t \leftarrow \sum_{j=r}^{N'} c_j \binom{j}{r} [\mathbf{a}^{j-r}]_t, \quad r=0, \dots, N'.$$

Mask usage policy: A fresh independent mask must be sampled for each aggregation round. Reusing the same mask across multiple rounds would leak inter-round differences: $\hat{\mathbf{x}}^{(\tau_1)} - \hat{\mathbf{x}}^{(\tau_2)} = \mathbf{x}^{(\tau_1)} - \mathbf{x}^{(\tau_2)}$, violating privacy. However, the same mask $\mathbf{a}$ is safely reused across different exponents *within* a single round, since only one value $\hat{\mathbf{x}} = \mathbf{x} - \mathbf{a}$ is opened and $\mathbf{a}$ remains hidden in the t -sharings.

Online phase (single opening): Given the sharings $\{[\text{sign}(\mathbf{g}_i)]_t\}_{i=1}^n$, the protocol proceeds as follows:

Step 1: Aggregation. Each party locally computes the aggregated sharing:

$$[\mathbf{x}]_t \leftarrow \sum_{i=1}^n [\text{sign}(\mathbf{g}_i)]_t.$$

This step requires no communication since it is a linear combination of local shares.

Step 2: Mask and open.

- 1) Compute $[\hat{\mathbf{x}}]_t \leftarrow [\mathbf{x}]_t - [\mathbf{a}]_t$
- 2) Open $[\hat{\mathbf{x}}]_t$ to obtain public value $\hat{\mathbf{x}} = \mathbf{x} - \mathbf{a}$

This is the only opening in the online phase, revealing the masked value $\hat{\mathbf{x}}$ which is statistically independent of $\mathbf{x}$ due to the uniform randomness of $\mathbf{a}$.

Step 3: MV polynomial evaluation. All users locally compute:

- 1) Public powers: $\hat{\mathbf{x}}^0, \hat{\mathbf{x}}^1, \dots, \hat{\mathbf{x}}^{N'}$
- 2) Output sharing:

$$[F(\mathbf{x})]_t = \sum_{r=0}^{N'} \hat{\mathbf{x}}^r \cdot [P_r]_t \quad (9)$$

This step requires no communication since $\hat{\mathbf{x}}^r$ are public values and $[P_r]_t$ are precomputed sharings.

The overall protocol yields a correct degree- t sharing of $F(\mathbf{x})$ within one online round with a single opening. Algorithm 6 summarizes the complete procedure.

Algorithm 6 Single-Mask Power Sharing for MV Polynomial Evaluation (Method B)

- 1: **Input:** User sharings $\{\text{sign}(\mathbf{g}_i)\}_t^n$ for local gradient of user i ; MV polynomial $F(\mathbf{x}) = \sum_{j=0}^{N'} c_j \mathbf{x}^j$
- 2: **Output:** Degree- t sharing $[F(\mathbf{x})]_t$
- 3: **Offline (input-independent; $t < n/2$ required for degree reduction):**
- 4: Sample fresh random mask $[\mathbf{a}]_t$
- 5: Initialize: $[\mathbf{a}^0]_t \leftarrow [1]_t$, $[\mathbf{a}^1]_t \leftarrow [\mathbf{a}]_t$
- 6: **for** $j = 2$ to N' **do**
- 7: $[\mathbf{a}^j]_{2t} \leftarrow [\mathbf{a}^{j-1}]_t \cdot [\mathbf{a}]_t$
- 8: $[\mathbf{a}^j]_t \leftarrow \text{DegRed}([\mathbf{a}^j]_{2t})$ {DN degree reduction}
- 9: **end for**
- 10: Precompute: $[P_r]_t \leftarrow \sum_{j=r}^{N'} c_j \binom{j}{r} [\mathbf{a}^{j-r}]_t$ for $r=0, \dots, N'$
- 11: **Online (per aggregation round):**
- 12: **Step 1: Aggregation (linear; no communication)**
- 13: $[\mathbf{x}]_t \leftarrow \sum_{i=1}^n [\text{sign}(\mathbf{g}_i)]_t$
- 14: **Step 2: Mask and open (single opening)**
- 15: $[\hat{\mathbf{x}}]_t \leftarrow [\mathbf{x}]_t - [\mathbf{a}]_t$
- 16: Open $[\hat{\mathbf{x}}]_t$ to obtain public value $\hat{\mathbf{x}} = \mathbf{x} - \mathbf{a}$
- 17: **Step 3: MV polynomial evaluation (local; no communication)**
- 18: Compute public powers: $\hat{\mathbf{x}}^0, \hat{\mathbf{x}}^1, \dots, \hat{\mathbf{x}}^{N'}$
- 19: Compute output sharing:

$$[F(\mathbf{x})]_t \leftarrow \sum_{r=0}^{N'} \hat{\mathbf{x}}^r \cdot [P_r]_t$$

- 20: **Return:** $[F(\mathbf{x})]_t$
-

Correctness: It is shown by Theorem 2.

Lemma 1 (Binomial Expansion). *For any $\mathbf{x} \in \mathbb{F}_p$ and mask $\mathbf{a} \in \mathbb{F}_p$ with $\hat{\mathbf{x}} = \mathbf{x} - \mathbf{a}$:*

$$\sum_{r=0}^k \binom{k}{r} \hat{\mathbf{x}}^r \cdot [\mathbf{a}^{k-r}]_t = [\mathbf{x}^k]_t.$$

Proof. By the binomial theorem, $(\hat{\mathbf{x}} + \mathbf{a})^k = \sum_{r=0}^k \binom{k}{r} \hat{\mathbf{x}}^r \mathbf{a}^{k-r}$. Since $\hat{\mathbf{x}} + \mathbf{a} = \mathbf{x}$, the result follows by linearity of sharing. $\square$

Theorem 2 (MV Polynomial Evaluation Correctness). *Algorithm 6 correctly computes $[F(\mathbf{x})]_t$.*

Proof.

$$\begin{aligned} [F(\mathbf{x})]_t &= \sum_{j=0}^{N'} c_j [\mathbf{x}^j]_t = \sum_{j=0}^{N'} c_j \sum_{r=0}^j \binom{j}{r} \hat{\mathbf{x}}^r [\mathbf{a}^{j-r}]_t \quad (\because \text{Lemma 1}) \\ &= \sum_{r=0}^{N'} \hat{\mathbf{x}}^r \sum_{j=r}^{N'} c_j \binom{j}{r} [\mathbf{a}^{j-r}]_t = \sum_{r=0}^{N'} \hat{\mathbf{x}}^r [P_r]_t. \end{aligned}$$

$\square$

Remark:

- **Security:** The single opening $\hat{\mathbf{x}} = \mathbf{x} - \mathbf{a}$ is one-time padded by the uniformly random mask $\mathbf{a}$, ensuring statistical privacy for $t < n/2$. Since $\mathbf{a}$ is uniformly distributed

and independent of $\mathbf{x}$, the opened value $\hat{\mathbf{x}}$ reveals no information about the aggregated input $\mathbf{x}$.

- **Complexity:** $O(N')$ offline complexity (storage and DN multiplications) with single-round online execution requiring only one opening—compared to Method A's $O(2^{N'})$ offline complexity with $O(N')$ online openings.
- **Structure:** Unlike Method A's two-layer architecture (DN multiplication followed by N -BTE combine), Method B employs a simpler single-layer structure: one linear masking step followed by local polynomial evaluation. This structural simplicity arises from exploiting the single-base property of MV polynomials.
- **Applicability:** Method B is specifically designed for MV polynomial evaluation where all multiplications involve powers of a single aggregated value $\mathbf{x}$. For general N -input multiplication, Method A remains necessary.

C. FL Framework via Secure Multiplication

In this section, we introduce a novel FL framework that performs secure aggregation through efficient secure multiplications to preserve user privacy while maintaining aggregation correctness. To clarify the internal mechanism of the proposed framework, we first describe the update procedures by the users and the central server, respectively.

1) User Update Procedure:

Step 1 (Sign Gradient Calculation): Each user i computes the local gradient $\mathbf{g}_i(\tau)$ using the global model $\theta(\tau)$ and performs 1-bit quantization to obtain the sign gradient:

$$\mathbf{x}_i(\tau) = \text{sign}(\mathbf{g}_i(\tau)), \quad \mathbf{x}_i(\tau) \in \{-1, +1\}^d, \quad (10)$$

where d denotes the dimensionality of the global model.

Step 2 (Secure Evaluation of $F(\mathbf{x}(\tau))$): Each user i securely evaluates its share of the MV polynomial $F(\mathbf{x}(\tau))$, resulting in a t -sharing $[F(\mathbf{x}(\tau))]_t^i$, where $\mathbf{x}(\tau) = \sum_{i=1}^n \mathbf{x}_i(\tau)$. This can be accomplished using Method A (Algorithm 5) or Method B (Algorithm 6), with Method B preferred for its linear rather than exponential offline complexity. The computation reveals only the masked value $\hat{\mathbf{x}}(\tau) = \mathbf{x}(\tau) - \mathbf{a}$ while keeping individual inputs $\mathbf{x}_i(\tau)$ private. Finally, each user transmits its secret share $[F(\mathbf{x}(\tau))]_t^i$ to the server.

2) Model Aggregation Procedure: Using the received secret shares $\{[F(\mathbf{x}(\tau))]_t^i : i \in [n]\}$, the server reconstructs the aggregated result $\tilde{\mathbf{g}}(\tau)$ via MDS-code-based decoding and then broadcasts it to all users:

$$\text{Decoding: } F(\mathbf{x}(\tau)) = \text{Dec}(\{[F(\mathbf{x}(\tau))]_t^i : i \in [n]\}), \quad (11)$$

$$\text{Broadcasting: } \tilde{\mathbf{g}}(\tau) = F(\mathbf{x}(\tau)) = \text{sign}(\mathbf{x}(\tau)). \quad (12)$$

The overall aggregation procedure is summarized in Algorithm 7. Each user generates and sends a secret share of the MV polynomial $F(\mathbf{x})$, while the server aggregates the received shares as in Eq. (11) and broadcasts the resulting global direction $\tilde{\mathbf{g}}(\tau)$ to all users for model update.

VI. THEORETICAL ANALYSIS

A. Dropout Resilience and Error-Correction Capability

We consider n users, a collusion-privacy threshold t , up to e malicious corruptions (errors), and up to s stragglers or

Algorithm 7 Secure Majority Vote Aggregation via Proposed Secure Multiplication

```

1: Input: Initial model  $\theta_0$ , learning rate  $\eta$ , number of users
    $n$ , MV polynomial  $F(\mathbf{x}) = \sum_{j=0}^N c_j \mathbf{x}^j$ 
2: for  $\tau = 0$  to  $T - 1$  do
3:   On User  $i$ :
4:     Compute local gradient  $\mathbf{g}_i(\tau)$ 
5:     Quantize gradient:  $\mathbf{x}_i(\tau) = \text{sign}(\mathbf{g}_i(\tau)) \in \{-1, +1\}^d$ 
6:     Generate secret share  $[F(\mathbf{x}(\tau))]_t^i$  via Algorithm 5 or 6
7:     Transmit  $[F(\mathbf{x}(\tau))]_t^i$  to the server
8:   On Server:
9:     Decode:  $F(\mathbf{x}(\tau)) = \text{Dec}(\{[F(\mathbf{x}(\tau))]_t^i : i \in [n]\})$ 
10:    Obtain majority vote:  $\tilde{\mathbf{g}}(\tau) \leftarrow F(\mathbf{x}(\tau))$ 
11:    Broadcast  $\tilde{\mathbf{g}}(\tau)$  to all users
12:   On User  $i$ : Update model:  $\theta(\tau + 1) \leftarrow \theta(\tau) - \eta \tilde{\mathbf{g}}(\tau)$ 
13: end for
14: Output: Final model  $\theta(T)$ 

```

dropouts (erasures). In the conventional DN-style binary multiplication, each multiplication doubles the sharing degree, i.e., t -sharing becomes $2t$ -sharing. The subsequent robust opening of $2t$ -sharing is performed via MDS-code-based decoding with minimum distance

$$d_{\min} = n - 2t. \quad (13)$$

A robust one-shot opening always succeeds whenever

$$2e + s \leq d_{\min} - 1 = n - 2t - 1, \quad (14)$$

where e denotes unknown errors (malicious corruptions) and s denotes erasures (dropouts with known locations). Due to the bound on error-correction capability in Eq. (14), the maximum correctable error and erasure counts are calculated as:

$$e_{\max} = \left\lfloor \frac{n - 2t - 1}{2} \right\rfloor \quad (s = 0), \quad s_{\max} = n - 2t - 1 \quad (e = 0).$$

In FL, adversarial fractions are typically small while dropout events are frequent. Choosing deliberately smaller thresholds $t \ll n/2$ (e.g., $t \leq n/4$ or $t \leq n/5$) enlarges the feasible decoding region, thereby improving dropout resilience and reducing the expected latency of robust openings. Reducing t by Δ increases $d_{\min}$ by 2Δ , directly expanding the correctable (e, s) region.

The implications of configuring t well below the classical $n/2$ bound (thus increasing $d_{\min}$ and improving tolerance to mixed errors and dropouts) are threefold: (i) stronger resilience against both malicious and straggling users, (ii) deterministic one-round success under design envelopes (e, s) , and (iii) amortized communication approaching the single-open baseline $O(n)$. Here, the “lower latency” effect refers to the reduced probability of decoding failure and thus fewer retries in robust openings, resulting in lower latency.

Design Rule: For deterministic one-round success, t must satisfy

$$t \leq \left\lfloor \frac{n - s_{\text{target}} - 2e_{\text{target}} - 1}{2} \right\rfloor, \quad (15)$$

while also meeting the privacy requirement $t \geq t_{\min}^{\text{privacy}}$. In practice, deployments often set t near the theoretical limit (e.g.,

$\lfloor (n-1)/2 \rfloor$ in the semi-honest model or $\lfloor (n-1)/3 \rfloor$ in the malicious model) to maximize collusion privacy. Nonetheless, deliberately smaller thresholds (e.g., $n/4$ or $n/5$) remain valid and enlarge $d_{\min}$, yielding stronger dropout resilience and lower expected latency while trading off collusion tolerance.

TABLE III: Error/erasure tolerance for selected t at $n=100$ (valid when $t < n/2$).

t	$d_{\min} = n - 2t$	$e_{\max}$ (if $s = 0$)	$s_{\max}$ (if $e = 0$)	Feasible (e, s)
49	2	0	1	$2e + s < 2$
45	10	4	9	$2e + s < 10$
40	20	9	19	$2e + s < 20$
33	34	16	33	$2e + s < 34$
25	50	24	49	$2e + s < 50$
20	60	29	59	$2e + s < 60$
15	70	34	69	$2e + s < 70$

As summarized in Table III, decreasing t proportionally increases the minimum distance $d_{\min} = n - 2t$, thereby enlarging the feasible region for error and erasure correction. This provides a clear design trade-off between privacy threshold, dropout resilience, and latency: smaller t strengthens error/erasure tolerance and reduces expected rounds, at the cost of reduced collusion privacy. This DN-style 2-input multiplication, where the sharing degree doubles after each multiplication, can be naturally generalized to higher fan-in settings in the first-layer design. More generally, a multiple-input multiplication maps t -sharing to Kt -sharing, where K represents the *degree-expansion factor* determined by the internal fan-in or grouping parameter.

B. Convergence Analysis

Theorem 3 (Convergence of SIGNSGD-MV under Errors and Erasures). *Consider SIGNSGD-MV, where up to e malicious corruptions (errors) and up to s stragglers or dropouts (erasures) may appear in each update round. Suppose the aggregation is performed using DN-based secure multiplication with MDS-code-based decoding of minimum distance $d_{\min}$. If the error-erasure condition $2e + s \leq d_{\min} - 1$ holds, then the effective aggregated gradient is equivalent to that of a non-adversarial, no-dropout environment.*

Under standard assumptions (smoothness, bounded variance, and unimodal symmetric gradient noise; see [14] for details), the convergence guarantees reduce to the benign case. Specifically, for $n_t = W^2$, we have

$$\left[\frac{1}{W} \sum_{i=0}^{W-1} \mathbb{E} \|\mathbf{g}_i\|_1 \right]^2 \leq \frac{4}{\sqrt{n_t}} \left[\frac{\|\boldsymbol{\sigma}\|_1}{\sqrt{M}} + \sqrt{\|\mathbf{L}\|_1 (f_0 - f^*)} \right]^2.$$

Sketch of Proof. The key observation is that MDS codes of minimum distance $d_{\min}$ can correct any pattern of e errors and s erasures provided that $2e + s \leq d_{\min} - 1$. Applying MDS-code-based decoding in the DN-based secure aggregation step ensures that the decoded value of each coordinate is identical to the true (non-adversarial) majority vote outcome whenever this condition is satisfied. Thus, the effective adversarial fraction becomes zero, regardless of the actual e or s within the decoding limit. By substituting $\alpha = 0$ into the convergence bounds established in [14], the theorem immediately yields

the results above. Therefore, as long as $2e + s \leq d_{\min} - 1$, the proposed secure aggregation achieves the same convergence behavior as in the benign setting without adversaries or dropouts. $\square$

C. Simulation-Based End-to-End Security

We analyze the end-to-end security of the proposed secure multiplication methods under the simulation-based paradigm, following the standard real/ideal execution framework in the semi-honest model with honest majority.

Security Definition: A protocol achieves *information-theoretic security* if, for every semi-honest adversary $\mathcal{A}$ corrupting at most t parties, there exists a simulator $\text{SIM}_{\mathcal{A}}$ such that the adversary's view is statistically indistinguishable from a simulated view given only the corrupted parties' inputs and the final output.

Security of Subprotocols:

Lemma 2 (Method A: DN + N-BTE). *Method A achieves information-theoretic security under $t < n/2$. All opened values in Layer 1 are one-time padded by uniformly random masks, and the precomputed subset sharings $\{\mathbf{a}_I\}_t$ in Layer 2 are never opened.*

Lemma 3 (Method B: Single-Mask Power Sharing). *Method B achieves information-theoretic security under $t < n/2$. The offline phase computes $\{\mathbf{a}^j\}_{j=0}^{N'}$ via DN multiplication, where all intermediate openings are one-time padded and input-independent. The online phase opens only $\hat{\mathbf{x}} = \mathbf{x} - \mathbf{a}$, which is statistically independent of $\mathbf{x}$ since $\mathbf{a}$ is uniformly random.*

End-to-End Security:

Theorem 4 (End-to-End Security for MV Polynomial Evaluation). *Under $t < n/2$, evaluating the MV polynomial via the proposed methods achieves information-theoretic security. Individual inputs $\{\mathbf{x}_i\}_{i=1}^n$ remain hidden; only the masked openings and the final output $F(\mathbf{x})$ are revealed.*

Sketch of Proof. The simulator $\text{SIM}_{\mathcal{A}}$ generates the adversary's view by sampling all opened values according to their input-independent distributions induced by one-time padding. Although the simulator receives the final output, it need not use it since every opened value can be sampled independently. The real and simulated views are statistically indistinguishable because every opened value is one-time padded: preprocessing-time openings (if any) are one-time padded and input-independent, and each online opening is likewise one-time padded (e.g., $\hat{\mathbf{x}} = \mathbf{x} - \mathbf{a}$), revealing no information about the underlying secrets. All remaining computations are performed locally on unopened sharings. Consequently, individual inputs $\{\mathbf{x}_i\}$ remain hidden as all opened values are statistically independent of the inputs. $\square$

Corollary 4.1 (Security of FL Framework). *Algorithm 7 achieves end-to-end information-theoretic security: individual sign gradients $\{\mathbf{x}_i(\tau)\}$ in each round τ remain hidden from the server and from any coalition of up to t users, who learn nothing beyond the final reconstructible aggregate. Fresh independent masks per round prevent cross-round leakage.*

D. Analysis of Per-User Online and Offline Costs

Table IV compares the per-user online and offline costs of the conventional Beaver-triple-based protocol [25] and the proposed methods across different scales. All costs are in bits, where one field element corresponds to $\lceil \log_2 p \rceil$ bits. Here, n denotes the number of users, p is the smallest prime greater than n , and $N' = (p-1)/2$ is the maximum polynomial degree after inverse-form reduction.

TABLE IV: Per-User Online and Offline Cost Comparison for Various Scales

Parameters	Method	Online Comm. (bits)	Latency (#rounds)	Offline Comp. (#mults)	Offline Storage (bits)
$n=10$ $p=11$	Beaver	144	4	9	108
	Method A	40 (−72.2%)	1 (−75.0%)	160	104
	Method B	8 (−94.4%)	1 (−75.0%)	4 (−55.6%)	24 (−77.8%)
$n=40$ $p=41$	Beaver	936	6	39	702
	Method A	240 (−74.4%)	1 (−83.3%)	2.10×10^7	6.29×10^6
	Method B	12 (−98.7%)	1 (−83.3%)	19 (−51.3%)	126 (−82.1%)
$n=60$ $p=61$	Beaver	1416	6	59	1062
	Method A	360 (−74.6%)	1 (−83.3%)	3.22×10^{10}	6.44×10^9
	Method B	12 (−99.2%)	1 (−83.3%)	29 (−50.8%)	186 (−82.5%)
$n=100$ $p=101$	Beaver	2772	7	99	2079
	Method A	700 (−74.7%)	1 (−85.7%)	5.63×10^{16}	7.88×10^{15}
	Method B	14 (−99.5%)	1 (−85.7%)	49 (−50.5%)	357 (−82.8%)

As shown in Table IV, Method A achieves 72–75% online communication reduction but incurs exponential offline cost, limiting its applicability to small N' . Method B achieves 94–99% online reduction with only one opening, while maintaining linear $O(N')$ offline complexity—over 50% computation and 77% storage reduction compared to Beaver-triple case.

Figure 2 visualizes these trends as n increases. Method B provides the optimal trade-off: single-round execution with minimal communication and linear offline complexity, making it the most practical choice for MV polynomial evaluation.

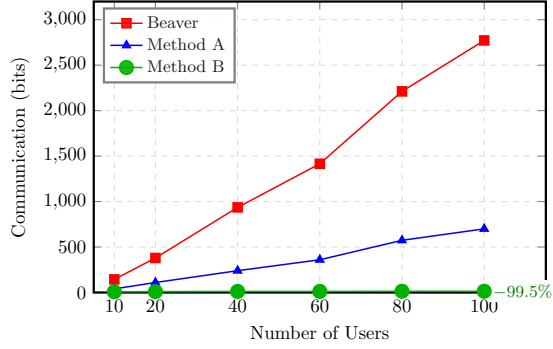
VII. EXPERIMENTS

A. Experiment Setup

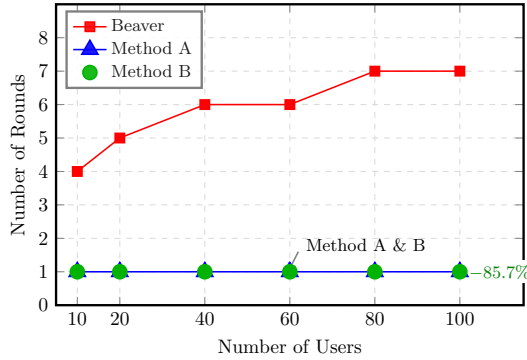
To evaluate the effectiveness and practicality of the proposed secure aggregation framework, we conducted extensive experiments on three benchmark datasets.

Datasets and Models. We validate the proposed method on three benchmark datasets: MNIST [51], FMNIST [52], and CIFAR-10 [53]. MNIST and FMNIST each contain 60,000 training samples and 10,000 test samples, while CIFAR-10 comprises 50,000 training samples and 10,000 test samples.

FL Configuration. We consider $N_t = 100$ users with non-IID data distributions, where each user is randomly assigned data from two classes following [42]. At each global round, a fraction $C \in [0.14, 0.21]$ of users is randomly selected to participate, resulting in $n \in [14, 21]$ active users per round. The privacy threshold is set to $t = \lfloor (n-1)/2 \rfloor$ to satisfy the honest-majority assumption. The prime p is chosen as the smallest prime greater than n , which determines the MV polynomial degree. With inverse-form exponent reduction (Section V-A), the effective degree is reduced to $N' = (p-1)/2$. Hyperparameter settings are summarized in Table V. All experiments were executed on a GPU server equipped with two NVIDIA RTX 3090 GPUs.



(a) Online communication cost.



(b) Online latency.

Fig. 2: Comparison of online performance: (a) communication cost and (b) latency.

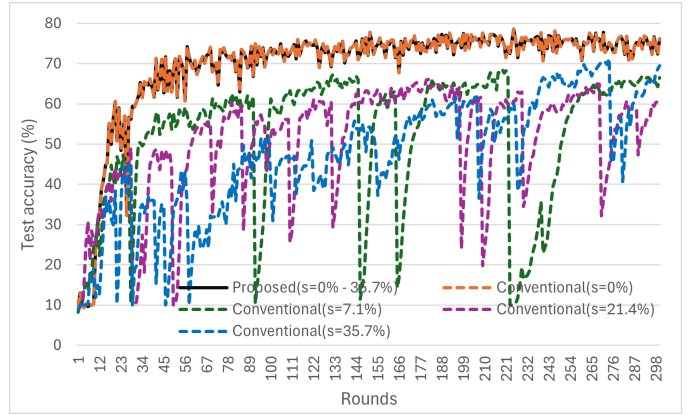
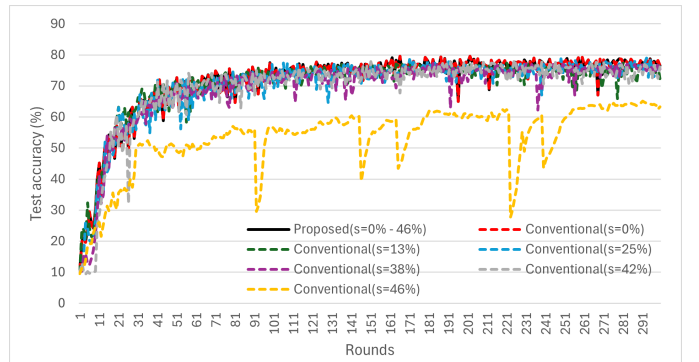
Evaluation Protocol. Since existing secure aggregation methods either lack dropout resilience or do not support adversary robustness within a unified framework, we compare against SIGNSGD-MV [14] and Hi-SAFE [25] as representative baselines. Note that both methods yield identical aggregation results since Hi-SAFE securely computes the same majority vote as SIGNSGD-MV; the key difference lies in privacy guarantees—Hi-SAFE provides computational security via Beaver triples, whereas the proposed method achieves information-theoretic security via single-mask power sharing (Method B). We therefore refer to these baseline methods collectively as “Conventional” in the following figures and tables.

TABLE V: Hyperparameter settings for all experiments.

Dataset	Learning Rate	Batch Size	Local Epochs
MNIST	0.001	100	1
FMNIST	0.005		
CIFAR-10	0.0005		

B. Experiment Results

We analyze the performance of the proposed framework under user dropout and adversarial settings. Unless otherwise stated, the honest-majority assumption ($t < n/2$) is adopted.

Fig. 3: Performance comparison under different dropout rates on the FMNIST dataset when $n = 14$.Fig. 4: Performance comparison under different dropout rates on the FMNIST dataset when $n = 21$.

1) *Dropout Resilience:* Figs. 3 and 4 illustrate the effect of user dropout on model accuracy when training on the FMNIST dataset under different numbers of participants ($n = 14$ and $n = 21$). As analyzed in Section VI-A, the proposed framework can tolerate up to $s_{\max} = n - 2t - 1$ erasures (dropouts) via MDS-code-based decoding. The proposed method exhibits identical accuracy to the baseline (i.e., the non-dropout case, $s = 0\%$) across all dropout rates, demonstrating complete resilience against missing or inactive users. In contrast, the accuracy of the conventional method degrades rapidly as dropout increases, especially when the number of active users becomes small.

TABLE VI: Test accuracy (%) under varying dropout rates on FMNIST ($n = 14$).

Dropout rate	$s = 0\%$	$s = 7.1\%$	$s = 21.4\%$	$s = 35.7\%$
Conventional	77.82	65.59	57.33	57.17
Proposed	77.82	77.82	77.82	77.82
Gain	—	+12.23	+20.49	+20.65

Table VI presents a detailed comparison of the average test accuracy between the proposed framework and the conventional method under various dropout rates on FMNIST with $n = 14$. While the conventional method suffers severe performance degradation from 77.82% to 57.17% as s increases to 35.7%, the proposed method consistently maintains 77.82%

accuracy regardless of dropout rate. Notably, despite providing end-to-end privacy protection via the proposed secure aggregation, the framework achieves up to 20.65 percentage-point improvement, demonstrating that it simultaneously ensures privacy preservation and dropout robustness even under severe participation variability.

2) *Adversary Robustness*: Figs. 5 and 6 show the performance under different adversarial rates on the MNIST and CIFAR-10 datasets. As established in Section VI-A, the MDS-code-based decoding can correct up to $e_{\max} = \lfloor (n-2t-1)/2 \rfloor$ errors. Under the honest-majority setting, the proposed method maintains the same accuracy as the baseline ($e = 0\%$) since MDS decoding enables correction of corrupted shares caused by adversarial users. Conversely, the conventional method exhibits steady degradation as adversarial behavior increases.

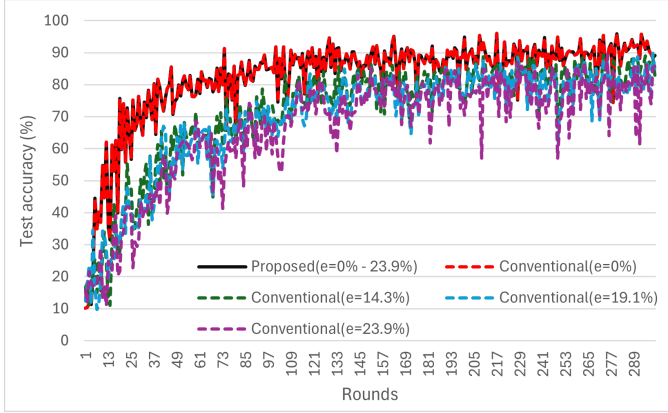


Fig. 5: Performance comparison under different adversarial rates on the MNIST dataset when $n = 21$.

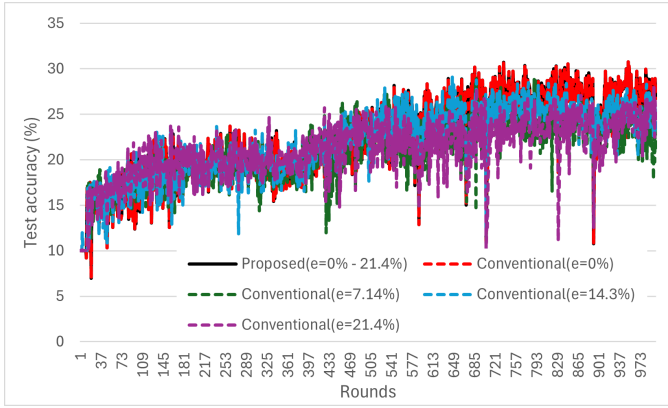


Fig. 6: Performance comparison under different adversarial rates on the CIFAR-10 dataset when $n = 14$.

TABLE VII: Test accuracy (%) under varying adversarial rates on MNIST ($n = 21$).

Adversarial rate	$e = 0\%$	$e = 14.3\%$	$e = 19.1\%$	$e = 23.9\%$
Conventional	90.02	83.60	81.93	79.28
Proposed	90.02	90.02	90.02	90.02
Gain	–	+6.42	+8.09	+10.74

Table VII provides a quantitative comparison of the proposed framework and the conventional method under adver-

sarial conditions on MNIST with $n = 21$. As the adversarial rate (e) increases, the conventional method loses up to 10.74 percentage-point accuracy, whereas the proposed method preserves perfect accuracy (90.02%) in all settings. Similar trends are observed on CIFAR-10 (Fig. 6), where the proposed method maintains stable accuracy while the conventional method degrades under adversarial conditions. These results demonstrate that the proposed approach remains fully robust to malicious updates while maintaining end-to-end privacy, confirming that the framework achieves both information-theoretic security and strong adversarial resilience.

VIII. CONCLUSION

This paper presented a lightweight and information-theoretically secure aggregation framework for sign-based FL on resource-constrained devices. Under the honest-majority assumption, the framework securely computes MV polynomials via single-round secure multiplication, revealing only the final aggregated sign without intermediate leakage. By leveraging MDS-code-based error correction, it achieves robustness against both user dropouts and adversarial behaviors. To enhance efficiency and scalability, we introduced two key techniques. First, inverse-form exponent reduction halves the effective MV polynomial degree, reducing both communication and computation costs. Second, we proposed single-round secure multiplication methods, where the structure-aware method exploits the single-base property of MV polynomials to achieve linear offline complexity and storage with only a single online communication. Together, these techniques reduce online communication by up to 99.5% and latency by up to 85.7% compared to conventional approaches. Experiments confirm that the framework maintains full accuracy under up to 35.7% dropout and 23.9% adversarial rates. Overall, the proposed framework establishes a practical foundation for large-scale, low-latency, and information-theoretically secure aggregation in sign-based FL. Future work will explore optimized implementations for heterogeneous edge environments and extensions to broader secure collaborative learning tasks.

APPENDIX A MULTI-ROUND EXTENSION OF METHOD A

While Method A achieves single-round online latency, its exponential offline cost $O(2^{N'})$ limits scalability for large polynomial degrees. A multi-round extension can distribute this workload across D_r sequential rounds.

Structure: The effective degree N' is divided into D_r stages, each handling at most $N'_{\text{round}} = \lceil N'/D_r \rceil$ inputs per stage. In each stage, partial products are computed and passed to the next stage. The feasibility constraint $t < n/2$ remains unchanged since all stages use degree- t sharings.

Complexity: The total offline storage decreases from $2^{N'} - N' - 1$ to approximately $D_r \cdot (2^{N'_{\text{round}}} - N'_{\text{round}} - 1)$, yielding exponential savings when $D_r > 1$. However, online latency increases from 1 to D_r rounds.

Remark: For MV polynomial evaluation, Method B's linear $O(N')$ complexity makes single-round execution practical without multi-round extensions. The parameter D_r serves as a design knob primarily for Method A when applied to general N -input multiplication scenarios.

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